

2. Problem Statement

Consider the following network configuration:

Network: A network with three hosts (h1, h2, h3) and two subnets (sA, sB). The network is configured as follows:

Configuration 5: The network is configured as follows: «Network configuration (Network configuration)».

Configuration: The network is configured as follows: Network configuration (Network configuration) reachable(h1,h2).

1. Network Configuration (Network configuration)

```
host(h1). host(h2). host(h3).
subnet(sA). subnet(sB).
host_in(h1,sA).
host_in(h2,sB).
host_in(h3,sA).
router(r1).
router_connects(r1,sA,sB).
link_up(h1,r1).
link_up(r1,h2).
firewall_allows(sA,sB,tcp).
```

R1: IF host_in(H1,S), host_in(H2,S), link_up(H1,_), link_up(H2,_)
THEN reachable(H1,H2).

R2: IF host_in(H1,S1), host_in(H2,S2), router_connects(R,S1,S2),
link_up(H1,R), link_up(R,H2), firewall_allows(S1,S2,Proto)
THEN reachable(H1,H2).

R3: IF reachable(H1,H2) THEN reachable(H2,H1).

R4: IF reachable(H1,Hx), reachable(Hx,H2) THEN reachable(H1,H2).

R5: IF firewall_blocks(S1,S2,Proto) ... THEN not_reachable(H1,H2).

2. Network Configuration (Network configuration)

```
WM0: { host_in(h1,sA), host_in(h2,sB), router_connects(r1,sA,sB),
link_up(h1,r1), link_up(r1,h2), firewall_allows(sA,sB,tcp) }
```

WM1: ... + reachable(h1,h2).

WM2: ... + reachable(h1,h2)

WM3: ... + reachable(h2,h1).

WM4: ... + reachable(h2,h1)

3. Network Configuration (Network configuration)

Goal: reachable(h1,h2)

R1 -> Network configuration (Network configuration).

R2 -> Network configuration (Network configuration), reachable(h1,h2) Network configuration.

4. Network Configuration (Network configuration)

Network configuration firewall_blocks(sA,sB,tcp) Network configuration (Network configuration) R2

R5. Network configuration: Network configuration (Network configuration), Network configuration (Network configuration) Network configuration

meta-override.